

Report 58 - Measurement, Control, and Reduced SMIB Assembly

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20 September 2026

Abstract

This report introduces the measurement and control layer required to close the hydropower frequency-control loop in OpenHPLjl. It adds scalar signal connectors, ideal frequency measurement, primary droop control, a first-order gate servo, PI control, a torque actuator, a classical rotor-angle model, and a single-machine infinite-bus network. These components are assembled into a reduced closed-loop SMIB for immediate simulation, while the full hydraulic-to-grid SMIB architecture is recorded as the next promotion target.

1 Updated Engineering Context

The core physical chain is

Reservoir → Waterway → Turbine → Shaft → Generator → Grid.

A runnable frequency-control study additionally needs

measurement → controller → actuator

around the plant. Hydropower transient-droop governors commonly act on guide-vane servomotors and use speed/frequency feedback [1]. Open power-system libraries likewise expose governor models with speed input, power/mechanical command output, lags, lead-lag blocks, and limiters [2, 3].

2 Generalized Concept Model

The closed loop is

$\omega \rightarrow \text{FrequencySensor} \rightarrow \text{DroopGovernor} \rightarrow \text{GateServo} \rightarrow \text{mechanical input} \rightarrow \text{shaft/generator/grid}.$

The first reduced SMIB is

Droop → Servo → Torque → Shaft → Classical Rotor → SMIB Network.

3 Measurement Components

A scalar `SignalPort` carries measured or commanded values.

For the ideal speed sensor,

$$y_\omega = \omega, \quad \tau_{\text{sensor}} = 0.$$

The zero-torque condition prevents the sensor from changing the mechanical dynamics.

4 Control Components

The primary droop law is

$$y_{\text{cmd}} = y_0 - \frac{\omega/\omega_{\text{ref}} - 1}{R}$$

so a positive frequency deviation closes the guide vanes.

The first-order servomotor is

$$T_g \dot{y} = y_{\text{cmd}} - y.$$

A generic PI controller is also provided:

$$e = r - y, \quad \dot{x} = e, \quad u = k_p e + k_i x.$$

5 Mechanical Actuation

The reduced SMIB uses a torque actuator

$$\tau_m = K_\tau y.$$

Because rotational-port torque is defined positive into a component, a torque source uses the opposite connector sign so the shaft receives positive driving torque.

6 Classical Generator and SMIB Network

Mechanical inertia is carried by the shaft, avoiding duplicated inertia.

The generator rotor angle follows

$$\dot{\delta} = \omega - \omega_s.$$

The classical lossless network is

$$P_e = P_{\max} \sin(\delta - \delta_g) + P_{\text{dist}}.$$

The generator consumes mechanical torque

$$\tau_e = \frac{P_e}{\omega}.$$

The shaft equation remains

$$J\dot{\omega} = \tau_m - \tau_e - D(\omega - \omega_0).$$

7 Reduced SMIB Implementation

The source assembly is

`src/Systems/ReducedSMIB.jl`

and contains

- LumpedShaft,
- FrequencySensor,
- DroopGovernor,
- GateServo,
- MechanicalTorqueActuator,
- ClassicalGeneratorRotor,
- SMIBNetwork.

The default electrical transfer limit is selected to make the initial operating point approximately balanced:

$$P_{\max} = \frac{K_\tau y_0 \omega_0}{\sin \delta_0}.$$

8 Numerical Experiment / Validation

The package test applies a nonzero electrical disturbance P_{dist} and solves the reduced SMIB for five seconds.

Validation checks include:

- successful ODE/DAE solution,
- finite shaft speed,
- finite rotor angle,
- finite guide-vane/servo state,
- nonzero frequency response,
- nonzero governor/servo response.

9 Full Hydro-SMIB Assembly

The next complete plant assembly is

Reservoir $\rightarrow$ RigidPipe $\rightarrow$ SurgeTank $\rightarrow$ Penstock
 $\rightarrow$ Controlled Turbine $\rightarrow$ Shaft $\rightarrow$ Generator $\rightarrow$ SMIB Grid

with

$$\omega \rightarrow \text{sensor} \rightarrow \text{governor} \rightarrow \text{servo} \rightarrow y.$$

The remaining technical tasks are:

1. promote turbine guide-vane opening from a parameter to a signal input;
2. expose turbine mechanical torque through `RotationalPort`;
3. resolve consistent hydraulic steady-state initialization;
4. add guide-vane position/rate limits;
5. add load/fault/step disturbance components;
6. validate the full assembly against a known operating point.

10 Expected Outputs

The reduced and later full SMIB should expose

$$f(t), \omega(t), \delta(t), P_m(t), P_e(t), y(t),$$

and, for the hydraulic plant,

$$Q(t), H(t), H_s(t), P_{\text{turbine}}(t).$$

11 Engineering Interpretation

This milestone separates the feedback architecture from component fidelity. The same measurement and governor components can later control a lookup-table Francis turbine, detailed waterway, and higher-order synchronous generator without redesigning the control loop.

Conclusion

OpenHPLjl now has the first measurement-control architecture required for an SMIB. The reduced SMIB is the immediate CI target. Once it is structurally stable, the same loop can be moved onto the complete nonlinear hydropower plant.

References

- [1] OpenHPL thesis resources: hydropower governing systems and transient droop. <https://build.openmodelica.org/Documentation/OpenHPL%203.0.1/Resources/Documents/Sharefi2011.pdf>
- [2] iPSL/OpenModelica GOVPSAT1 turbine-governor documentation. <https://build.openmodelica.org/Documentation/iPSL.Electrical.Controls.Eurostag.govpsat1.html>
- [3] iPSL/OpenModelica TGOV1 turbine-governor documentation. <https://build.openmodelica.org/Documentation/iPSL.Electrical.Controls.Eurostag.tgov1.html>